CBS

INSTRUCTIONS

GEK- 7842

IC4965, CONTROL PANEL FOR SCR DC DYNAMOMETER

G-E Requisition No.

INDUSTRY CONTROL DEPARTMENT
GENERAL ELECTRIC
SALEM, VA.

DESCRIPTION

IC4965 SCR D-C DYNAMOMETER CONTROL

The d-c dynamometer system is an adjustable voltage system using the d-c dynamometer as a conventional d-c shunt motor.

A motor-generator set consisting of a squirrel cage induction motor or a synchronous motor and a d-c generator are used in conjunction with the dynamometer as shown in Figure 1.

When the dynamometer is motoring, power is taken from the a-c line and supplied to the dynamometer through the motor-generator set. When the dynamometer is absorbing, power is returned to the a-c line.

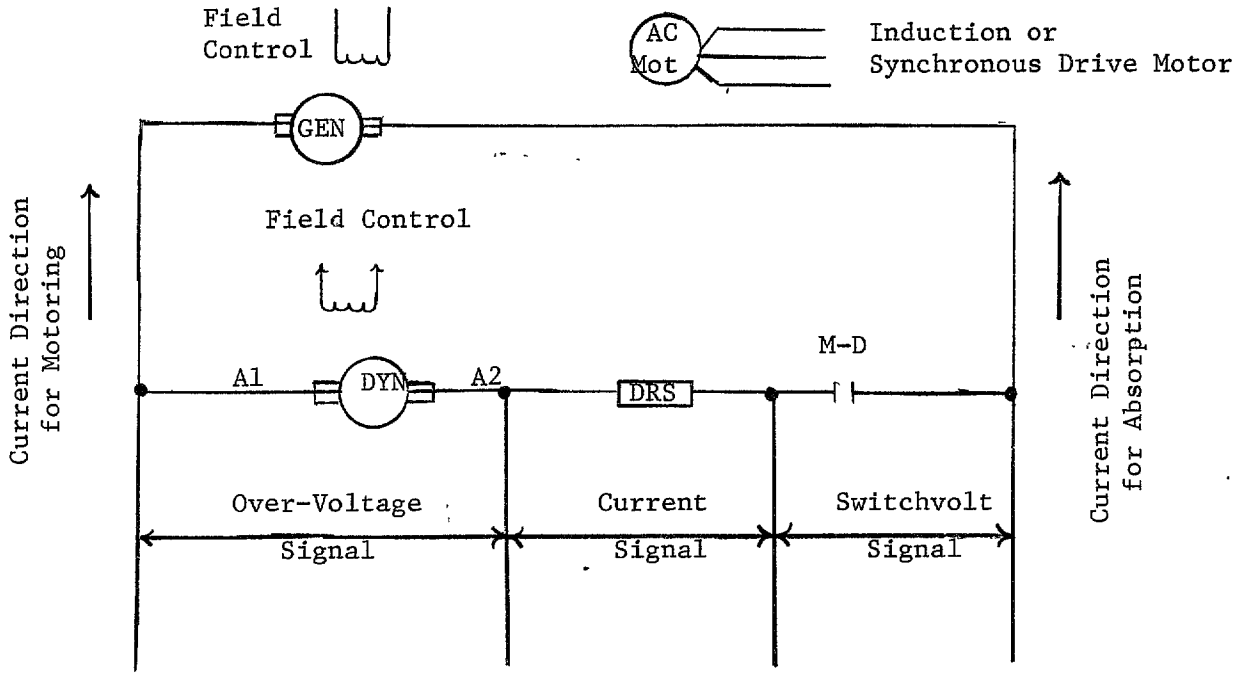
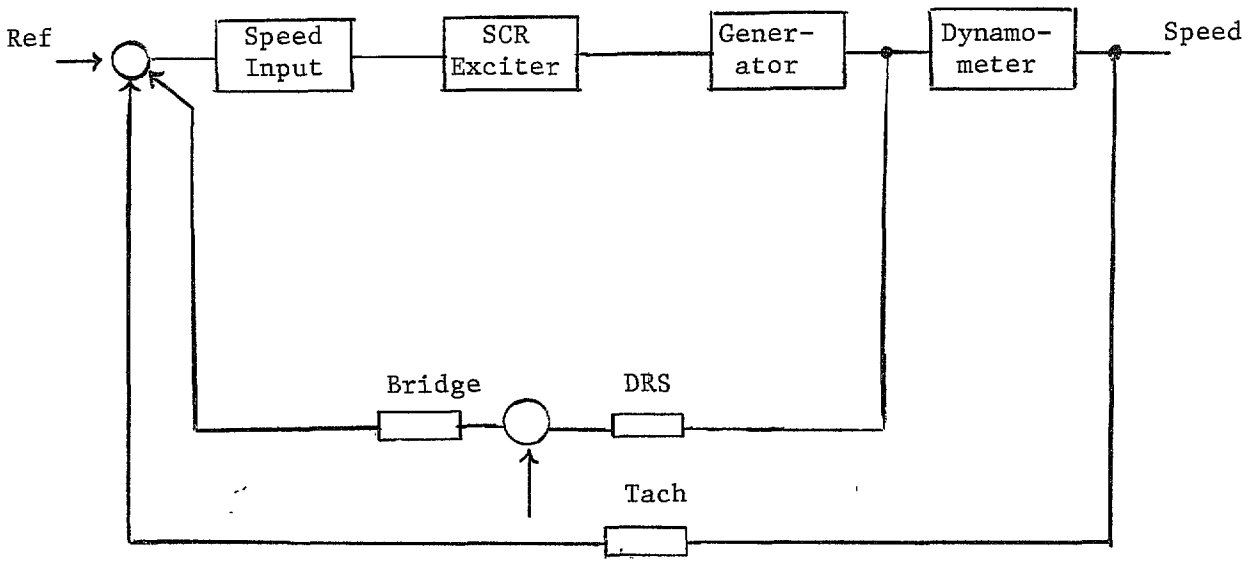


Figure 1

Control of the dynamometer and generator is accomplished by field control of both machines, using an SCR exciter for each. (See below for field adjustment.) A regulated power supply is used to obtain reference signals. Speed may be accurately controlled by use of suitable input circuits. A block diagram is shown in Figure 2.



ADJUSTMENT

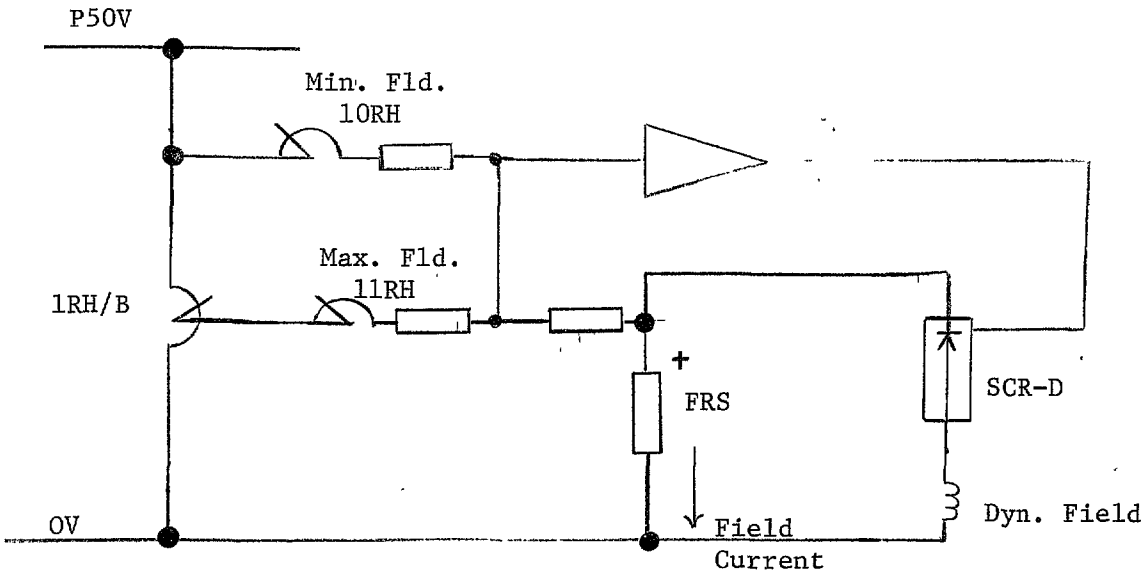
The proper procedure for adjusting the power supply, current limit circuits, stability circuits, maximum speed, dynamometer minimum field, dynamometer field current feedback, and overspeed relay are described later.

POWER SUPPLY

A regulated power supply is furnished in order to supply constant reference signals - minimizing the drift of the dynamometer operation. See IC3601A103 and IC3601A105 for details.

DYNAMOMETER FIELD CONTROL

The dynamometer field current is supplied by the dynamometer silicon controlled rectifier package SCR-D. This SCR unit serves as the dynamometer field current regulator. The desired base speed field current is set by rheostat 11RH (see Figure 3). The voltage drop across resistor FRS in series with the dynamometer field is compared with this voltage reference. The error signal is amplified by the preamplifier and SCR in order to maintain the desired dynamometer field current.



When a speed higher than base speed is desired, some method of changing the reference to the dynamometer field current regulator is needed. This tapering of the field is accomplished by shaping.

Adjust max. and min. field as follows:

10RH must be adjusted first with 1RH fully CW.
From the equation below, calculate the approximate setting for 10RH.

WF = Top Speed Field Amp
FF = Base Speed Field Amp

$$10RH = \frac{33K \times 50V}{WF \times FRS} - 220K$$

From the following equation, calculate the approximate setting for 11RH. When adjusting 11RH, 1RH must be fully CCW.

$$11RH = \frac{33K \times 50V}{(FF-WF)\ FRS} - 100K$$

A sample calculation for 10RH, 11RH from the given values:

FF = 10 Amperes
WF - 2.3 Amperes

$$10RH = \frac{33K \times 50V}{2.3} - 220K = 500K$$

$$11RH = \frac{33K \times 50V}{7.7} - 100K = 114K$$

Set 10RH, 11RH to calculated values and energize dynamometer field with loop open and generator stopped. Operate 1RH through its range and observe dynamometer field current to follow plot on sheet 1A. If end points of curve do not correspond with plot, adjust 10RH at WF and 11RH at FF to give rated conditions. Note that 10RH must be adjusted first.

OVERSPEED REALY ADJUSTMENT

Before attempting to operate the system at high speeds, the overspeed pot 2RH should be set at a low resistance for immediate shutdown in the event of overspeeding the dynamometer during setup. As higher speeds are desired, increase resistance of 2RH until top speed has been reached. 2RH should then be adjusted to pick up OS-D at 103% of rated top speed.

SPEED CONTROL

The system is capable of closely maintaining speed with 100 percent change in load torque. A d-c tachometer-generator, coupled to the dynamometer, supplies a voltage directly proportional to speed. This voltage is compared with a voltage reference set by a speed adjustment potentiometer on the operator's panel which is connected to the regulated reference supply as shown in Figure 4.

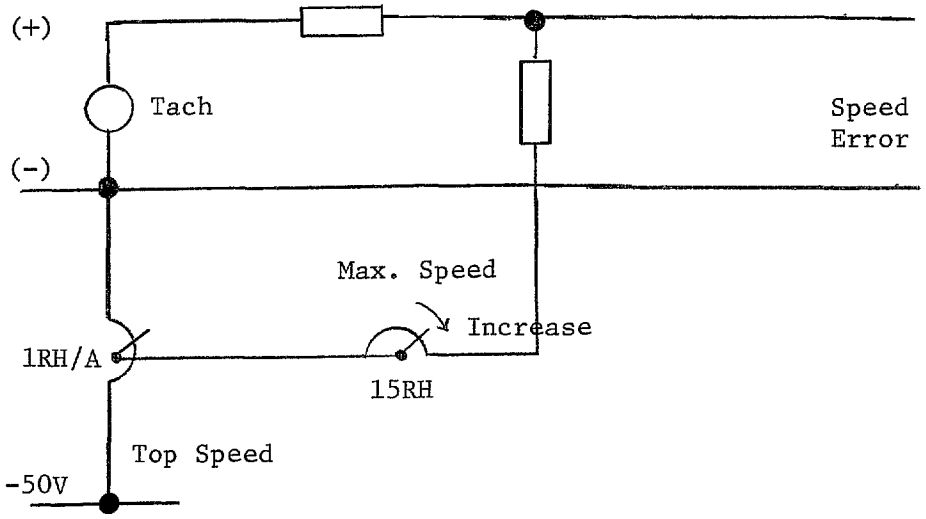


Figure 4 Speed Error Circuit

If the speed signal does not agreee with the reference, the resultant error voltage will be applied to the generator SCR to correct the speed.

The SCR (which will also be described later) excites the generator field. The generator changes the speed of the dynamometer (by changing its terminal voltage) in a direction to rebalance the tachometer-generator and reference signals.

To set maximum speed control:

1. Set 15RH full CCW (Resistance in).

2. Start system, dynamometer motoring, and set percent rotation of 1RH from curve on sheet 1A for base speed.

3. Adjust 15RH until base speed is reached.

4. Advance 1RH and check for correct top speed at full CW rotation. 15RH may require trimming at this point.

CURRENT LIMIT CIRCUIT

Maintaining speed within close limits requires a high gain system. This means that a small error between the tachometer generator voltage and the reference voltage will cause a large change in motor armature voltage. This large change could cause the armature current to become excessive. Therefore, a circuit which limits motor armature current is included to portect the machinery.

This is accomplished by modifying the speed error with a signal from the armature dropping resistor (see Figure 5).

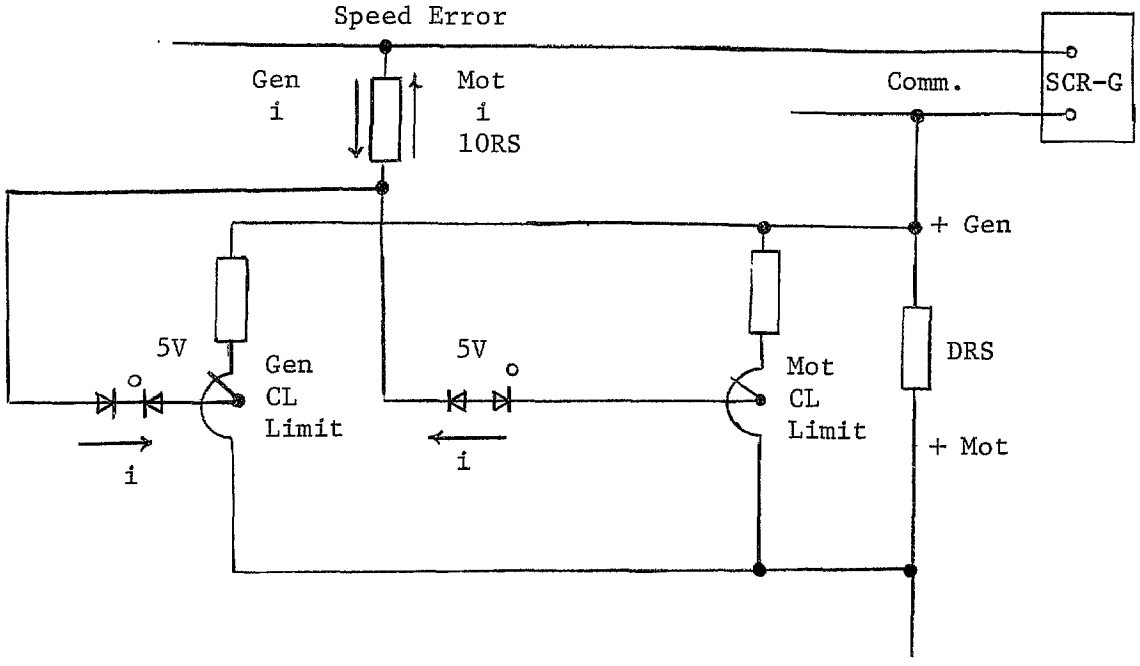


Figure 5

When the voltage of DRS causes the setting of the current limit pots to exceed 5 volts (Zener diode voltage is 5 volts), current will flow in the directions shown by the arrows. This will increase or decrease generator voltage as needed to lower the armature current. The range of current limit is 66% to 175% of full load current.

STABILIZING CIRCUIT

Again, the system must have a high gain in order to obtain the accuracy desired. In high-gain systems, where a feedback is compared with a reference signal, instability may become a problem.

Instability is caused by energy storage devices and components within the closed loop system - in particular the inertia of the dynamometer and test engines, but also the armature loop circuit inductance, and generator field inductance.

Stabilization is accomplished by use of a lag-lead or "notch" circuit (see Figure 6), which is the equivalent of the simple resistor-capacitor combination shown in Figure 7.

This circuit alters the effect of input signals according to the magnitude of their rate of change with respect to time.

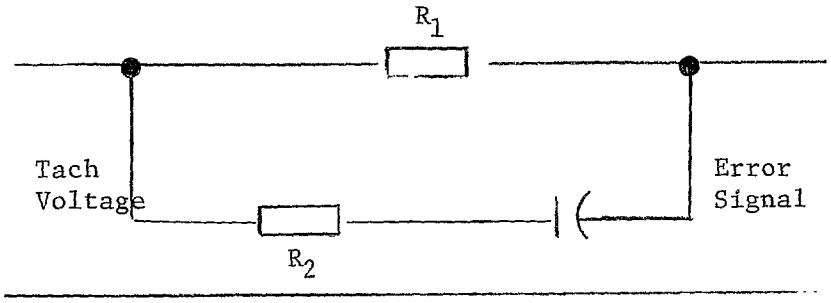


Figure 6

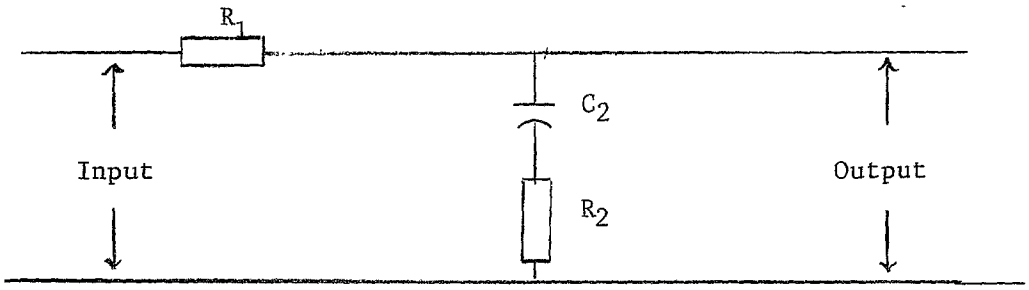


Figure 7

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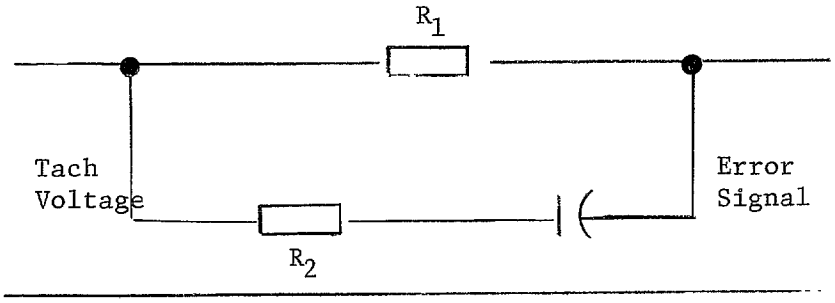


Figure 6

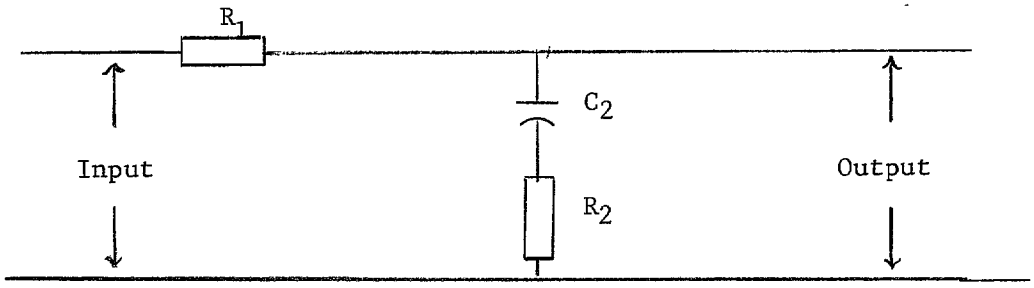


Figure 7

To adjust stability circuits, in the event of instability, two methods are provided.

1. Notch or lag-lead circuit is changed by adding more capacitance to C_2. (Figure 7) 10 MF is usually supplied in this circuit. To improve stability, add parallel capacitance. Do not add any more capacitance than necessary for satisfactory operation, or exceed 40 MF.

2. If the above method proves ineffective, then the forward gain of the system should be reduced. This is done by increasing resistance of gain pot 16RH until desired stability is reached.

For operation of the system above base speed of the dynamometer, the generator voltage is maintained approximately constant and the dynamometer field current is reduced. This is a constant horsepower type of operation, in contrast to the constant torque operation below base speed. As the dynamometer field current is lowered by the speed control potentiometer on the operator's panel when operating above base speed, the dynamometer tachometer-generator is still supplying a signal which is compared to the setting of the operator's speed control potentiometer. This error operates as before to adjust the generator field, hence dynamometer armature voltage. Therefore, regulation of speed is still accomplished by means of dynamometer armature voltage. Above base speed, the armature voltage is primarily a trimming action, in that main control is obtained by dynamometer field weakening. The operator must be careful not to exceed the horsepower rating of the system while above base speed.

INSTALLATION

Receiving and Unpacking

The control equipment is mounted in a rigid floor-mounted enclosure; and panels are crated and shipped in an upright position. When a panel is received, it should be handled from the top and kept upright.

A panel which is not to be installed immediately should be stored in a clean, dry location. When the panel is installed, the control should be unpacked carefully, making certain that no small parts are thrown away with the packing material.

Check the nameplate rating of the machine to be sure that it agrees with that given on the panel nameplate. The control is designed for use with specific equipment and must not be applied to any other without first checking with the nearest Apparatus Sales Office of the General Electric Company.

After the panel has been unpacked, it should be handled from the removable lifting angle on top provided for that purpose. To avoid undue strains on the panel, sling the lifting rope through the holes near the ends of the lifting angle.

Clean the inside of the control panel with a brush, soft cloth, or dry compressed air. All dirt, dust, or bits of packing material, which interfere with successful operation of the panel devices, should by all means be removed from the panel. You should likewise take care during the cleaning operation to prevent any dirt from being blown into the inaccessible spaces of the devices.

Installing the Panel

When the panel is to be installed, it is essential that it be securely fastened in a true, upright position on a level surface to permit proper operation of the devices mounted on the panel.

The enclosing case is mounted on two floor sills, one each across the back and front of the enclosing case. Floor sills consist of channel irons which are four inches deep and one and three-quarters inches high. The four-inch dimension is horizontal. On single-case installations the floor sills may be removed and the enclosing case bolted on the floor. On lineups of more than one case it is recommended that floor sills be used to give rigidity to the panel. Mounting holes for one-half-inch bolts are provided in each sill. The outline for the particular panel to be installed should be checked to determine the location of the mounting holes. It is recommended that the one-half-inch mounting bolts be grouted in the floor in the proper location for fastening the panel.

There are several possible methods of installing the floor sills. The most frequently used methods are described below:

If the panel will be installed on an existing floor, grout the one-half-inch anchor bolts in the floor and slip the panel over them. Check level position of the sills with a machinist's level, and shim the sills if necessary. Check that the surfaces of the front and back sills are in a plane, so the angle iron supports of the panel are plumb. If the panel is installed in this manner, there will be a two-inch gap between the floor and sides of the panel enclosure. This gap should be closed to prevent accumulation of dirt under the panel, which in due time may interfere with the proper operation of the panel devices.

If the concrete floor will be poured at the time that the panel is installed, grout the one-half-inch anchor bolts in the sub-floor, and slip the panel over them. Check alignment of the floor sills with a level and make sure that the panel supports are plumb. Then pour concrete around the floor sills. This should be done carefully, so as not to splash concrete on the panel devices. Care should also be taken to finish the surface of the concrete so there will be no interference of the enclosing case door; and so there will be a tight joint between the floor and the enclosure side, thus preventing the entrance of dirt at the bottom of the panel.

If the concrete floor is to be poured before the panel will be installed, a dummy set of floor sills can be used. These dummy sills can be made of wood of the proper size and length. The dummies should be set in place, together with the one-half-inch anchor bolts, before the concrete is poured, and then replaced with the actual sills, after the panel is received.

Provisions should be made for incoming and outgoing cable and control wires before the panel is in place. In all cases the outline for the panel being installed should be used to determine the space available, and the panel interconnection diagram should be checked for the number of wires and cables which will be installed. The cables normally enter the bottom of the panels.

Preparing the Panel for Operation

Before you can operate the panel even for a tryout without power, all devices must be placed in fully operative condition. The armatures of some of the relays and contactors may have been tied or blocked to prevent damage during shipment. Remove all of these ties and blocks before operating devices.

Sealing surfaces of unplated and laminated a-c contactor and relay magnets, as well as the ends of shafts, are coated with a rust preventative. This protection against rust is a heavy grease which does not interfere with the proper operation of the devices, but which encourages the accumulation of dust and dirt which may ultimately prevent proper device operation.

Therefore, before operating the panel, you should wipe off all rust preventative. Magnet sealing surfaces may then be coated with a thin film of light machine oil.

Operate each device and relay by hand to see that the moving parts operate freely without binding. Make sure that all electrical interlocks are clean and make good contact when closed. Relays and contactors are carefully adjusted and tested at the factory. If adjustment information for individual devices is needed, however, it may be found in individual instructions for each device.

With the power leads disconnected at the machine terminals, but will all devices properly readied for operation, apply power and operate the panel to make certain that its operation is in agreement with the elementary diagram.

The leads may then be connected and the equipment put into operation.

MAINTENANCE

All electrical apparatus will provide maximum trouble-free service if given the benefit of preventative maintenance inspections and periodic cleanings. Although the frequency of the inspection periods will depend on operating conditions, it is important that a definite inspection routine and inspection schedule be followed. Again, maintenance instructions in the individual device instructions are most helpful.

Renewal Parts

Order renewal parts by part number and description, and identify the equipment for which they are required. Structural parts and complete operating devices are identified on the equipment drawing list. They are available on request.